SOFC PLANT · 50 kW PROOF-OF-CONCEPT

How the SOFC plant works, and which devices are on in each stage

A plain-language walkthrough of the system layout with sensors. Redrawn and simplified; updated 2026-11-02.

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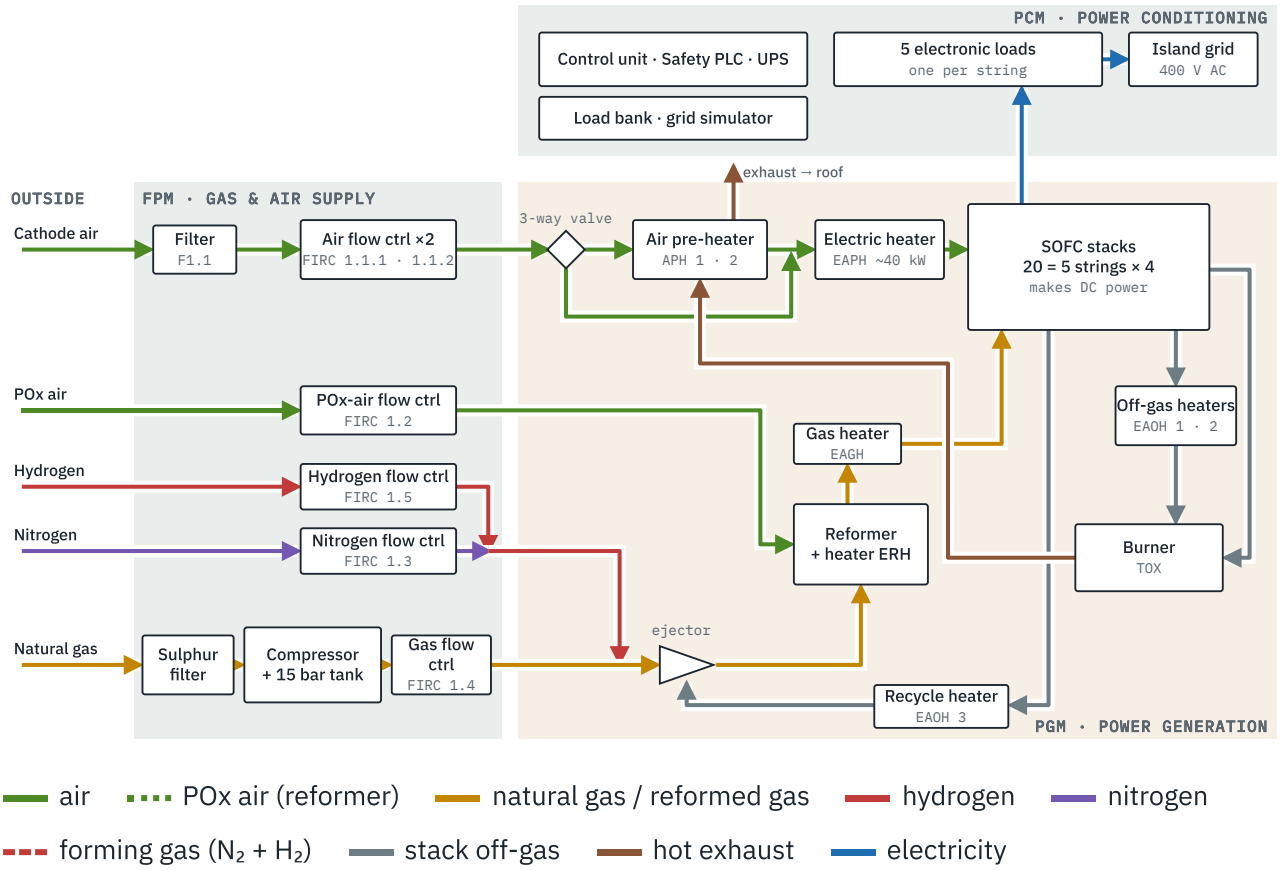
SOFC PLANT · 50 KW PROOF-OF-CONCEPT · HOW IT WORKS

Gas and air go in, electricity and warm exhaust come out.

A solid oxide fuel cell (SOFC) is a ceramic “battery” that never runs flat. It makes electricity as long as it gets fuel on one side and air on the other, and it only works when it is very hot, around 600 °C.

> Everything else on the drawing exists to **feed the stacks** (clean, measure and heat the air and gas), to **use what the stacks leave behind** (burn the left-over fuel and reuse its heat), and to **take the electricity away safely**.

The plant in four zones



Simplified redraw of the plant. Press Play to see the order in which things flow: air, then fuel, then the stack off-gas, then exhaust, then electricity.

Outside the container

`5 supplies`

Cathode air, POx air, hydrogen, nitrogen and natural gas arrive from site pipes.

FPM · supply module

`clean · reduce pressure · meter`

Filters, pressure reducers, shut-off valves and one flow controller per gas. Natural gas is de-sulphured and compressed into a 15 bar tank.

PGM · power generation

`heat · reform · react · burn`

Air heaters, reformer, the 20 stacks, the burner and heat exchangers. This is the hot box.

PCM · power conditioning

`take the power away`

Five electronic loads turn stack DC into 400 V AC for an island grid. Also the control unit, safety PLC and UPS.

The journey in six steps

1. **Air** is filtered, metered and pre-heated, first by the hot exhaust, then by a ~40 kW electric heater, and enters the cathode (air side) of the stacks. Page 2
2. **Natural gas** is de-sulphured, compressed, metered, mixed with recycled off-gas and a little air, and turned into a hydrogen-rich gas in the **reformer**. Page 3
3. In the **stacks** oxygen from the air passes through the ceramic and reacts with the hydrogen. That reaction pushes electrons through the wires: electricity.
4. Gas the stacks did not use is partly **recycled** to the reformer and the rest is **burned** in the burner (TOX). Page 4
5. The burner's hot exhaust **pre-heats the incoming air** and then leaves through the roof.
6. The DC power goes through five **electronic loads** to the island grid, watched by voltage and current sensors and by the safety PLC. Page 5

How to read the sensor bubbles

Each round bubble on the drawing is one sensor. Its letters say what it measures and what is done with the reading. On the drawing temperatures are yellow, pressures green, safety trips red.

LETTER	MEANS	EXAMPLE ON THE DRAWING
T	Temperature	`TIR 2.05` air temperature before the electric heater
P	Pressure	`PIR 2.02` air pressure at the stack inlet
F	Flow	`FIRC 1.4` natural-gas flow
E	Electrical (voltage, current, power)	`EIR 4.1.10–4.1.21` stack voltages of string 1
L / Q	Level / quality	`LIR 7.01` water level in the condensate trap
…I R	Indicated and Recorded (shown and logged)	`TIR` = temperature, shown and logged
…C	Controlled: a controller uses it to act	`TIRC 2.06` the heater is driven to hit this temperature
…Z	Safety trip: crossing a limit stops the plant	`TIRZ 8.1.3` stack above 720 °C → emergency purge

Which machines are running depends on the plant stage (heat-up, standby, fuel switch-over, part load, cooling). Page 6 lights them up stage by stage.

IF YOU REMEMBER ONE THING

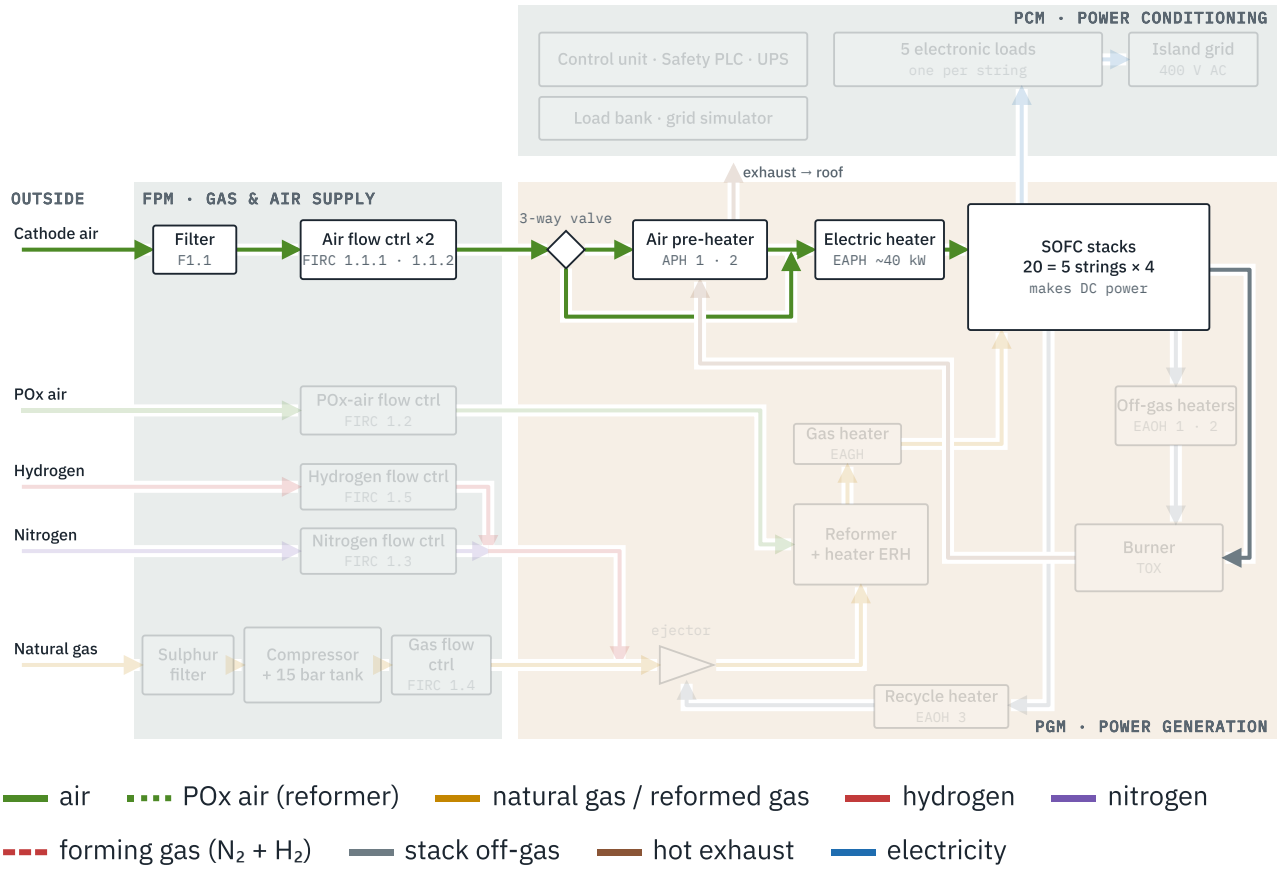
The stacks make the power; every other device on the drawing heats, feeds, cleans up after or protects them.

Redrawn and simplified from the project P&ID “System layout with sensors” (50 kW PoC).
Tags as on the drawing; sensor meanings from the project instrument map.

PAGE 2 · CATHODE AIR

Air is the stack's oxygen supply and its main heater and cooler.

The stacks need oxygen on their air side (the cathode). The same air stream carries heat in during start-up and carries heat out during operation and cooling.



Air path highlighted. The 3-way valve sends part of the air through the pre-heaters and lets the rest bypass them, to set the temperature.

Step by step

1. **Filter** `F1.1` with a humidity and oil indicator keeps dirt, water and compressor oil out.
2. **Pressure reducers** `PR 1.1.1 · 1.1.2` drop the site air from 6 to 5 bar.
3. **Two air lines in parallel**, each with a shut-off valve `MV 1.1.2 · 1.1.3` and a flow controller `FIRC 1.1.1 · 1.1.2`, up to 6000 litres per minute each.

4. **3-way valve** `TWV` : splits the air between the pre-heaters and a bypass. It is steered by temperature, not by flow.
5. **Air pre-heaters** `APH 1 · 2` : heat exchangers. The hot burner exhaust warms the air without mixing with it. Free heat once the plant runs.
6. **Electric air heater** `EAPH` , about 40 kW. Does almost all the heating during start-up and trims the temperature later.
7. **Stack air inlet**. The air then leaves the stacks warmer and slightly poorer in oxygen and goes to the burner (page 4).

Sensors on this path

SENSOR	WHERE	WHY IT MATTERS
`PIR 1.1.1 · 1.1.3`	Before each air flow controller	Enough supply pressure?
`FIRC 1.1.1 · 1.1.2`	Air flow controllers	How much air. Higher air flow = faster cooling.
`TIR 1.1.1` `PIR 1.1.2`	After the two lines merge	Air leaving the supply module
`FIRC 1.1.3`	3-way valve	Valve controller. Its reading equals `TIR 2.05` (it is temperature-controlled).
`TIR 2.04 · 2.1.1 · 2.1.2`	Into the pre-heaters	Cold-side inlet temperatures
`TIR 2.02 · 2.03`	Out of the pre-heaters	How much heat the exhaust gave
`TIR 2.05` `PIR 2.01`	Before the electric heater	Starting point for the heater
`EIRC 2.01`	Electric heater	Heater power (kW)
`TIRC 2.06`	After the electric heater	The temperature the heater is controlled to = stack air-inlet temperature
`TIRZ 8.1.1`	Electric heater	Safety trip above 1050 °C

SENSOR	WHERE	WHY IT MATTERS
`PIR 2.02`	Stack air inlet	Air pressure at the stack. Shows air is flowing.
`TIRC 5.1.1 · 5.1.2` `PIR 5.1.1 · 5.1.2`	Stack air outlet	Temperature and pressure of the air leaving the stacks

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IF YOU REMEMBER ONE THING

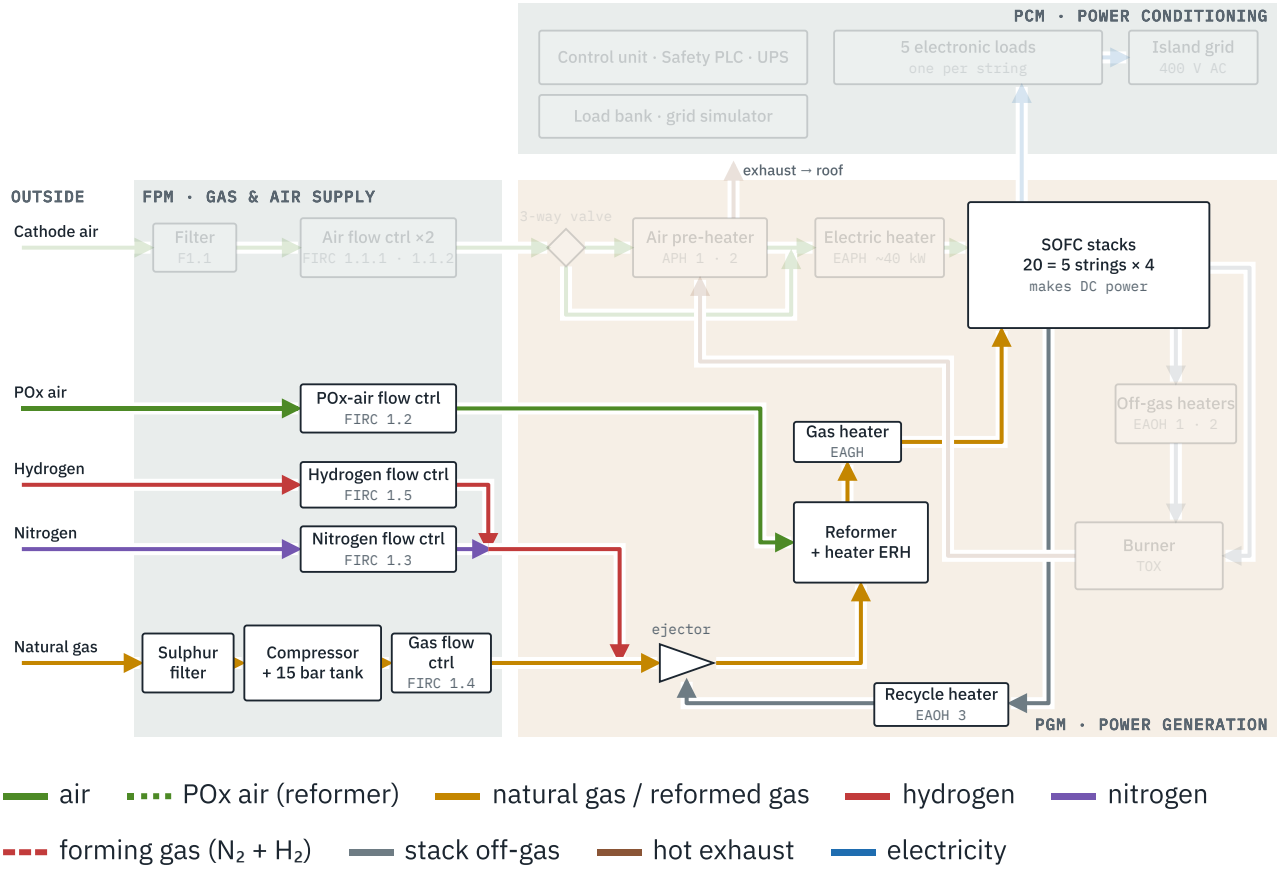
Air is filtered, metered, warmed twice (by exhaust heat, then electrically) and then enters the stacks; the electric heater’s outlet temperature TIRC 2.06 is the key setting.

Redrawn and simplified from the project P&ID “System layout with sensors” (50 kW PoC).
Tags as on the drawing; sensor meanings from the project instrument map.

PAGE 3 · FUEL SIDE

Natural gas is cleaned, compressed and turned into hydrogen before it reaches the stacks.

The stacks cannot use natural gas (mostly methane) directly. A reformer turns it into a hydrogen-rich gas. During start-up and shut-down a safe “forming gas” of nitrogen with a little hydrogen is used instead.



Fuel side highlighted: natural gas (amber), POx air (dotted green), forming gas (dashed red) and the recycle loop through EAOH 3.

Natural gas, step by step

1. **Supply** arrives at low pressure (20–30 mbar). Shut-off valves `MV 1.4.1 · 1.4.2` close it on demand.

2. **Sulphur filter** (desulphuriser): two cartridge beds plus dust filters. Sulphur would poison the reformer and the stacks.
3. **Compressor + 15 bar storage tank**: a 3-stage compressor fills a tank. A relief valve protects it.
4. **Pressure reducer** `PR 1.4` 15 → 13 bar, then shut-off valve `MV 1.4.3` and the **gas flow controller** `FIRC 1.4` (0–200 litres per minute).
5. **Ejector**: a jet pump. The fast natural-gas stream sucks in recycled stack off-gas, which brings steam and heat back to the reformer. No moving parts.
6. **Reformer** with its electric heater `ERH`: methane + steam → carbon monoxide + hydrogen. A small stream of **POx air** `FIRC 1.2` burns a little of the gas inside to help with heat.
7. **Gas heater** `EAGH` keeps the reformed gas hot, then it enters the stack fuel side (the anode).

Forming gas: nitrogen + hydrogen

Nitrogen `FIRC 1.3` and hydrogen `FIRC 1.5` come in on their own lines (pressure 10 → 5 bar) and join the fuel line. Hydrogen keeps the stack fuel side chemically protected while it is hot and no natural gas flows. Nitrogen alone is used to flush (purge) the lines.

Sensors on this path

SENSOR	WHERE	WHY IT MATTERS
`QIR 1.4` `PIRZ 8.1.2`	Gas supply	Gas indicator; low-pressure safety trip
`EIC 1.4`	Compressor	Ready / running / alarm signals
`PIR 1.4.1`	After the 15 → 13 bar reducer	Tank pressure is enough
`FIRC 1.4`	Gas flow controller	Natural gas flow. Runs only before and during power.
`TIR 1.4.1 · 1.4.2` `PIR 1.4.2`	Leaving the supply module	Gas temperature and pressure
`FIRC 1.2` `PIR 1.2.1 · 1.2.2`	POx air	Air for the reformer
`FIRC 1.3 · 1.5` `PIR 1.3.1 · 1.5`	Nitrogen, hydrogen	Forming-gas flows and supply pressures

SENSOR	WHERE	WHY IT MATTERS
`TIR 3.01 · 3.02` `PIR 3.01`	Into the ejector	Gas entering the hot box
`TIR 6.01` `PIR 6.01`	Recycled off-gas into the ejector	Recycle loop is working
`TIRC 3.03` `PIR 3.02`	Ejector outlet	Mixed gas
`TIR 3.04 · 3.05 · 3.06`	Reformer inlets	POx air and mixed-gas temperatures
`TIRC 3.07 – 3.16`	Inside the reformer bed	Catalyst temperatures (3.07 inlet, 3.15 outlet)
`TIRC 5.17` `TIRZ 8.2.5`	Reformer heater ERH	Control; trip above 860 °C
`TIRC 3.17`	Reformer outlet	Reformed gas
`TIRC 3.1.1` `TIRZ 8.2.1`	Gas heater EAGH	Control; trip above 950 °C
`TIR 3.18 · 3.19.1 · 3.19.2` `PIR 3.03`	Stack fuel inlet	Gas arriving at the stacks

IF YOU REMEMBER ONE THING

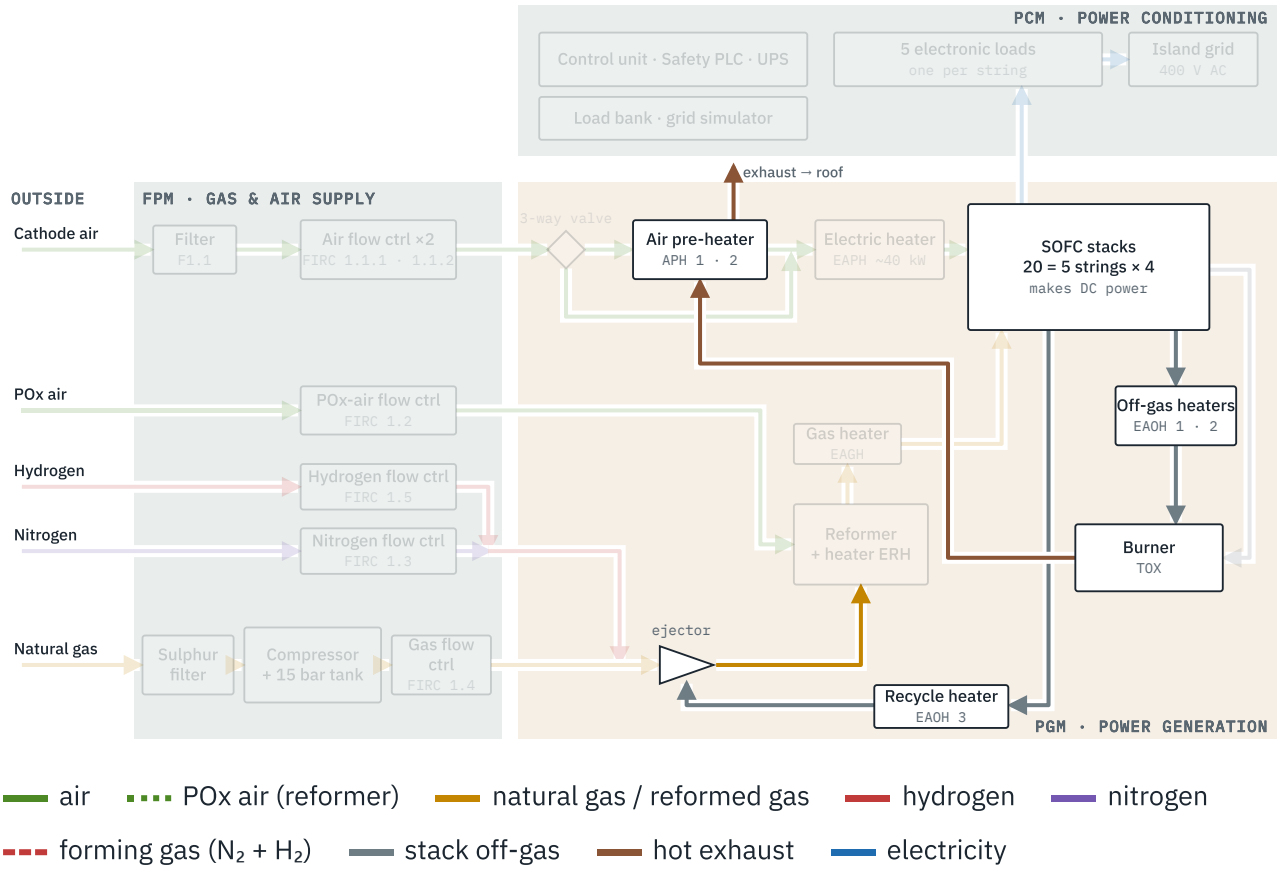
The reformer is the translator: it turns natural gas into the hydrogen the stacks can use, and a jet pump recycles stack off-gas to give it the steam it needs.

Redrawn and simplified from the project P&ID “System layout with sensors” (50 kW PoC).
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PAGE 4 · AFTER THE STACK

Nothing is thrown away hot: left-over fuel is burned and its heat warms the new air.

The stacks use most, not all, of the fuel. What comes out of the fuel side still contains hydrogen and carbon monoxide. Part of it goes back to the reformer; the rest is burned.



Off-gas (grey) and exhaust (brown) highlighted. The exhaust passes the air pre-heaters before it leaves through the roof.

Step by step

1. **Fuel-side off-gas** leaves the stacks `TIRC 5.2.1 · 5.2.2`.
2. **Recycle branch**: heater `EAOH 3` keeps it hot on the way back to the ejector (page 3).
3. **Burner branch**: heaters `EAOH 1 · 2` keep it hot on the way to the burner, so no water condenses.
4. **Air-side off-gas** from the stacks also goes to the burner, bringing the oxygen to burn with.

5. **Burner (TOX)**: hydrogen + carbon monoxide + oxygen → carbon dioxide + water. Runs at roughly 500–600 °C.
6. **Air pre-heaters** `APH 1 · 2` : the exhaust gives its heat to the incoming air (page 2).
7. **Exhaust** leaves through the roof. Water that condenses is collected in a trap with a level sensor `LIR 7.01` and drained `V7.01` .

Sensors on this path

SENSOR	WHERE	WHY IT MATTERS
`TIRC 5.2.1 · 5.2.2` `PIR 5.2.1 · 5.2.2`	Stack fuel outlet	Off-gas leaving the stacks
`TIRC 6.1.1` `TIRZ 8.2.4`	Recycle heater EAOH 3	Control; trip above 950 °C
`TIRC 5.3.1 · 5.3.2` `TIRZ 8.2.2 · 8.2.3`	Off-gas heaters EAOH 1 · 2	Control; trip above 950 °C
`TIRC 5.5.x` `PIR 5.4.x` `TIR 5.4.x` `PIR 5.3.x`	Burner inlets (fuel and air sides)	What enters the burner
`TIRC 5.07 – 5.15`	Inside the burner	Burner temperatures (5.09 is the main one)
`TIRZ 8.1.9`	Burner	Safety trip above 930 °C
`TIR 5.6.1 · 5.6.2`	Air-side bypass	Air that skips the burner
`TIR 7.1.x · 7.2.x · 7.3.x` `PIR 7.1.x`	Through the pre-heaters	Exhaust in and out = heat handed to the air
`TIR 7.04 · 7.05` `PIR 7.02 · 7.03`	Exhaust line	Final exhaust
`TIRZ 8.1.10`	Exhaust outlet	Safety trip above 300 °C
`LIR 7.01`	Condensate trap	Water level

IF YOU REMEMBER ONE THING

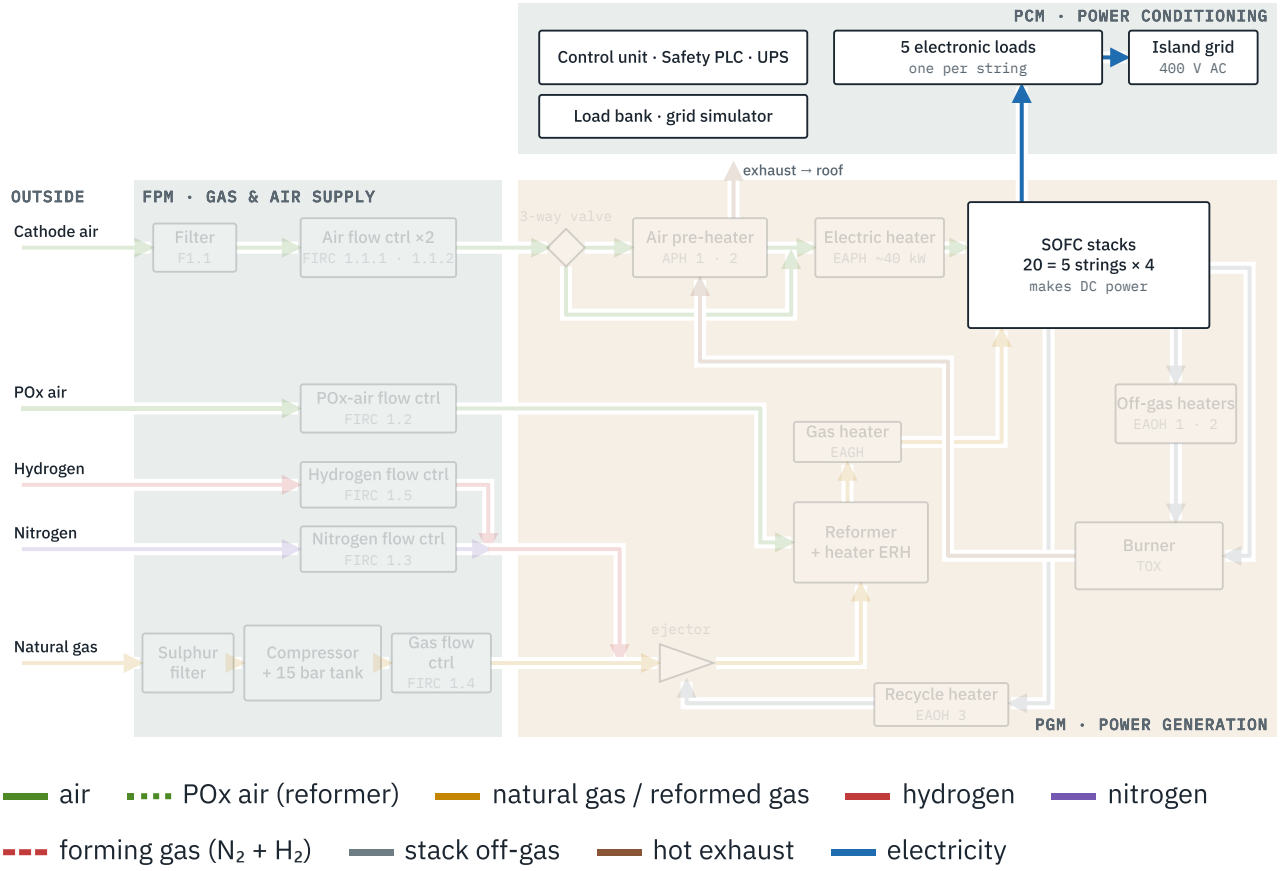
Stack off-gas is either recycled to the reformer or burned; the burner’s heat goes back into the incoming air, so the plant needs little electric heating once it runs.

Redrawn and simplified from the project P&ID “System layout with sensors” (50 kW PoC).
Tags as on the drawing; sensor meanings from the project instrument map.

PAGE 5 · ELECTRICITY, CONTROL AND SAFETY

Twenty stacks in five strings feed five electronic loads, all under a safety controller.

Each string is four stacks in a row. Each string has its own electronic load, which decides how much current is drawn and turns it into 400 V AC for an island grid (a small local grid, not the public one).



Electrical side highlighted. Design rating 60 kW DC / 50 kW AC, 30 A per string.

The parts

Stacks

EIR 4.s.10 – 4.s.21 · TIR 4.6.1–4.6.36

Voltage of every stack and stack half in each string; 36 temperatures inside the stack

Electronic loads ×5

EIRC 4.s.1 – 4.s.9

One per string, 14 kW each. Read string voltage and current; set how much current

module. Safety trip `TIRZ 8.1.3` above 720 °C.

to draw. Insulation trip `BE 1.1 – 5.1` below 2 kΩ.

Island grid · load bank

`EIRC 9.x · 9.7`

400 V AC, 50/60 Hz. A resistor load bank and a grid simulator absorb the power for testing.

Control unit

`normal control`

Runs the controllers (heaters, flows, valves) and logs every sensor.

Safety PLC

`all …Z sensors`

Separate controller. If a trip sensor stays past its limit for more than 50 ms, it purges or stops the plant on its own.

UPS

`EIRC 9.6`

Battery backup so control and safety keep working if site power fails.

What the safety PLC does on a trip

TRIP TYPE	CAUSED BY	WHAT HAPPENS
Emergency **purge**	Any temperature trip: EAPH above 1050 °C, stack above 720 °C, burner above 930 °C, exhaust above 300 °C, EAGH / EAOH 1–3 above 950 °C, ERH above 860 °C. Stack insulation below 2 kΩ. Pressure PIRZ 8.1.2 below 750 mbar absolute (location not yet confirmed). Loss of exhaust-fan air flow.	Air valves open (2 × 3,000 Nl/min), nitrogen (1,200 Nl/min) and hydrogen (60 Nl/min) flush the fuel side, natural gas and POx air close, both gas vents open. Heaters, stacks and loads disconnect; compressor stops.
Emergency **stop**	Gas detector: O_2 below 17 %, H_2 above 40 % LEL, CO above 60 ppm. Smoke. Emergency-stop button.	All supply valves close, air and forming gas included; only gas vent MV 1.4.4 opens. Heaters, loads and compressor switch off. No purge: the plant goes straight to **shutdown** (page 6).

`IF YOU REMEMBER ONE THING`

The loads decide how much power is taken from each string, and a separate safety controller can shut everything down by itself.

Redrawn and simplified from the project P&ID “System layout with sensors” (50 kW PoC).
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PAGE 6 · WHICH DEVICES ARE ON

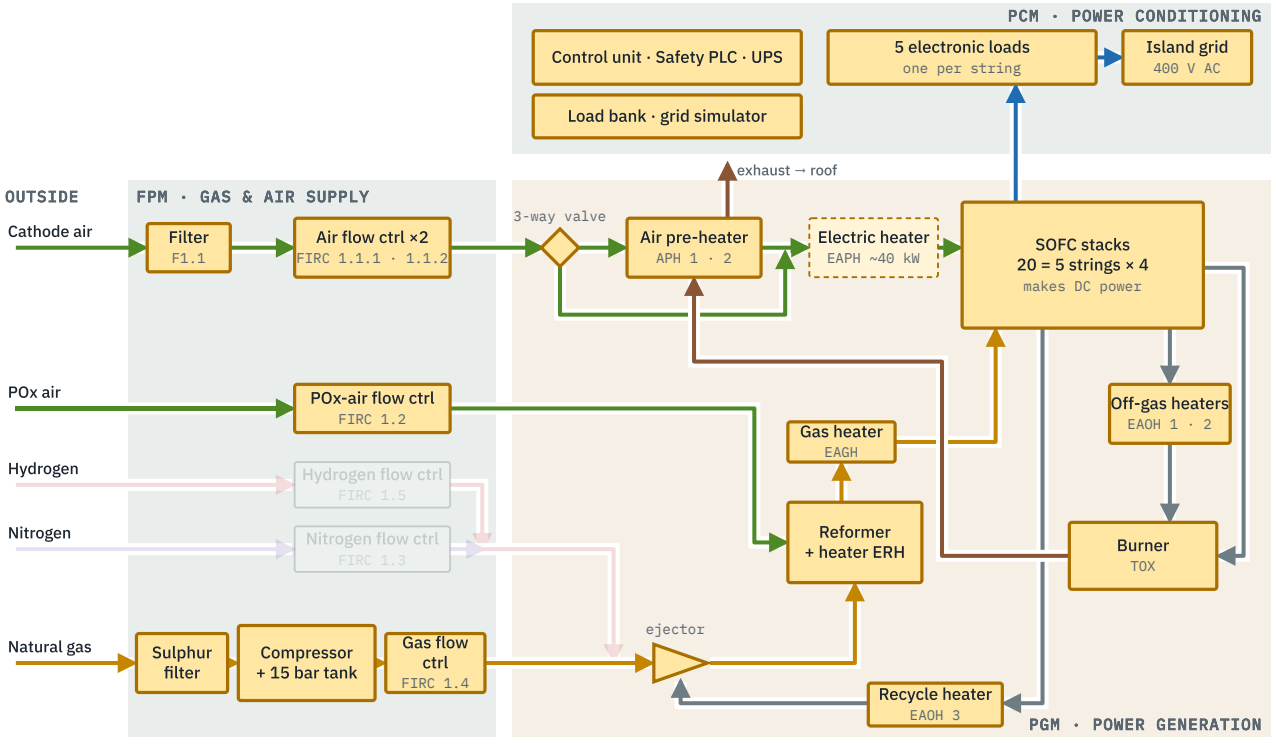
What is switched on depends on the stage the plant is in.

Pick a stage. Devices that run light up; dashed ones run only partly, at low power, or work passively (they only do something when hot gas flows through them). Faded devices are off.

Heat-up air | Heat-up N_2+H_2 | Hot standby | Fuel switch | **Part load**

Air-flow cooling | Shutdown | Trip: purge

Part load. Seen up to 34 kW; rated power (50 kW AC) not yet reached. Exhaust heat now pre-heats the air. At the end the load ramps to zero. (seen in the logged data)



air · POx air (reformer) · natural gas / reformed gas · hydrogen · nitrogen · forming gas (N_2 + H_2) · stack off-gas · hot exhaust · electricity

Lit = on. Dashed = partly on, low power or passive. Faded = off. The control unit, safety PLC and UPS are always on.

All stages at once

● on ◌ partly / low / passive ○ off

DEVICE	HEAT-UP AIR	HEAT-UP N_2+H_2	HOT STANDBY	FUEL SWITCH	PART LOAD	AIR-FLOW COOLING	SHUTDOWN	TRIP: PURGE
Air supply (filter, valves, 2 air flow controllers)	●	●	●	●	●	●	○	●
3-way valve (air temperature split)	◌	◌	◌	◌	●	◌	○	○
Air pre-heaters APH 1 · 2 (passive: heat whatever hot gas passes)	●	●	●	●	●	●	○	○
Electric air heater EAPH	●	●	◌	◌	◌	○	○	○
Nitrogen line	○	●	●	◌	○	●	○	●
Hydrogen line	○	●	●	◌	○	●	○	●
Natural gas: sulphur filter, gas flow ctrl	○	○	○	●	●	○	○	○
Compressor + 15 bar tank	○	○	◌	●	●	○	○	○
POx air to reformer	○	○	○	●	●	○	○	○
Ejector (jet pump)	○	◌	◌	●	●	○	○	○
Reformer + heater ERH	●	●	●	●	●	○	○	○
Gas heater EAGH	●	●	●	●	●	○	○	○

DEVICE	HEAT-UP AIR	HEAT-UP N_2+H_2	HOT STANDBY	FUEL SWITCH	PART LOAD	AIR-FLOW COOLING	SHUTDOWN	TRIP: PURGE
Off-gas heaters EAOH 1 · 2	●	●	●	●	●	○	○	○
Recycle heater EAOH 3	●	●	●	●	●	○	○	○
Burner TOX	○	○	○	◌	●	○	○	○
Stacks making power	○	○	○	○	●	○	○	○
Electronic loads	○	○	○	○	●	○	○	○
Island grid	○	○	○	○	●	○	○	○
Control unit · Safety PLC · UPS	●	●	●	●	●	●	●	●

The pattern in plain words

- **Air and the control system** run in every stage except shutdown. Air is how heat goes in and out.
- **The electric air heater** works hardest during start-up (~13 kW with air only, ~20 kW with forming gas). Once hot, it trims at ~10 kW; in air-flow cooling and shutdown it is off.
- **Forming gas (N_2 + H_2) follows one rule:** it is on whenever the stack is **above 210 °C** and no natural gas flows, to protect the fuel side from air. That alone covers heat-up N_2+H_2, hot standby, the emergency purge and the hot part of air-flow cooling, so no stage-by-stage list is needed. Only an emergency stop overrides it (everything closes). In the 39 logged days the rule matches 96 % of the minutes.
- **The small gas-line heaters** (ERH, EAGH, EAOH 1–3) run from the first start-up step until part load ends, keeping the fuel pipes hot. They have a short **switch-on delay**: they come on 2–3 minutes after the air starts (at most 9 min), and a few hot-standby periods start up to 13 min late. Those first minutes still belong to the stage; the heaters are shown on because they run for the rest of it. A heater counts as on even when it runs below full power: before natural gas they draw about two-thirds of their top current.
- **The burner** shows no measurable burning on forming gas; it lights during the fuel switch-over.

- **Natural gas and POx air** run only in the fuel switch-over and part load; the compressor starts refilling its tank late in hot standby.
- **The two safety trips** differ: an emergency purge keeps air and forming gas flowing to protect the hot stack; an emergency stop closes everything, air included, so the plant is simply in shutdown.
- **Stacks and loads** carry current only in the “part load” stage, which always follows a fuel switch-over after 5–30 minutes.
- **After part load** comes air-flow cooling (~7 °C per hour), then shutdown with everything off, where the stack cools by itself (under 1 °C per hour).

Basis: the gas phases, heater power, air flow and current seen in the 39 logged days and the project stage rules; the trip-purge column and the emergency stop inside shutdown follow the safety functions in the component sensor map (section 4). The gas-line heaters (ERH, EAGH, EAOH 1–3) are measured by their current sensors IIR 5.1.3, 3.1.1, 5.1.1, 5.1.2 and 6.1.1; the minutes before they switch on (switch-on delay, up to 15 min) are not counted as off. Burner burning is judged from its outlet versus inlet temperature.

IF YOU REMEMBER ONE THING

Cold plant: air and electric heat. Hot but idle: air plus forming gas. Part load: natural gas, reformer, burner and loads, about 10 minutes after the gas is switched on. Air-flow cooling: air (with forming gas while still hot). Shutdown (also straight after an emergency stop): everything off.

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